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Where Minds Pollinate

Class 10

Biology

Chapter 7

Short Questions Notes

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Chapter 7 – Inheritance

Section II: Short Answer Questions

1. Why do chromosomes need to compact into thick chromatids during cell division?

During cell division, chromosomes condense into thick, compact structures called **chromatids** to ensure that DNA is divided accurately between daughter cells. This compaction prevents the long DNA strands from tangling or breaking and allows them to move efficiently along the spindle fibers during mitosis and meiosis. Compact chromatids make the process of genetic material separation organized and error-free.

2. How does the structure of telomeres help protect genetic information during cell division?

Telomeres are repetitive DNA sequences found at the ends of chromosomes. They act like protective caps, preventing chromosomes from sticking to each other or losing important genes during replication. Every time a cell divides, telomeres shorten slightly, but this controlled shortening helps protect the main genetic code. Without telomeres, DNA damage and loss of genetic information would occur, leading to cell malfunction or ageing.

3. How do Mendel's experiments with pea plants support the idea of dominant and recessive traits?

Mendel's pea plant experiments showed that when two plants with contrasting traits were crossed, only one trait appeared in the first generation (F_1), while the other was hidden. When the F_1 plants were self-pollinated, the hidden trait reappeared in the F_2 generation in a **3:1 ratio**. This proved that some traits are **dominant** (visible) and others **recessive** (hidden but not lost), forming the foundation of Mendel's **Law of Dominance**.

4. Why did Mendel use true-breeding plants in his experiments?

Mendel used **true-breeding plants** because they consistently produced offspring with the same traits generation after generation. This ensured that the plants were **genetically pure** and carried identical alleles for a given trait. By using these plants, Mendel could accurately study how traits were passed from one generation to the next without any hidden variations interfering with his results.

5. How can two parents with free earlobes have a child with attached earlobes?

Free earlobes (E) are dominant over attached earlobes (e). Two parents with free earlobes can have a child with attached earlobes if both parents are **heterozygous (Ee)**. In this case, there's a 25% chance that the child will inherit two recessive alleles (ee), resulting in attached earlobes. This happens because each parent can pass either the dominant or recessive allele to their child.

6. How does the law of segregation ensure genetic diversity in offspring?

Mendel's **Law of Segregation** states that during gamete formation, the two alleles for a trait separate, so each gamete carries only one allele. When fertilization occurs, the combination of alleles from two parents produces unique genetic combinations. This random mixing of alleles increases **genetic diversity**, making every individual genetically different, except identical twins.

7. Why did Mendel's dihybrid cross results differ from his monohybrid cross results?

In a **monohybrid cross**, Mendel studied one trait and found a **3:1 ratio** of dominant to recessive traits in the F₂ generation.

In a **dihybrid cross**, he studied two traits at once (for example, seed colour and shape) and found a **9:3:3:1 ratio**. This showed that genes for different traits are inherited **independently** of each other — a principle known as **Mendel's Law of Independent Assortment**.

8. In pea plants, the allele for round seeds (R) is dominant over wrinkled seeds (r). If a heterozygous round-seeded plant (Rr) is crossed with a homozygous wrinkled-seed plant (rr), what will be the genotypic and phenotypic ratios of offspring?

Cross: $Rr \times rr$

- **Gametes:** R, r and r, r
- **Offspring Genotypes:** 50% Rr (round), 50% rr (wrinkled)
- **Genotypic Ratio:** 1 Rr : 1 rr
- **Phenotypic Ratio:** 1 Round : 1 Wrinkled

Thus, half of the offspring will have round seeds, and half will have wrinkled seeds.

9. If a true-breeding tall pea plant (TT) is crossed with a short pea plant (tt), what will be the genotype and phenotype of the F₁ generation? What will be the genotypic and phenotypic ratios of the F₂ generation if two F₁ plants are crossed?

- **P Generation Cross:** TT × tt
- **F₁ Generation:** All Tt (Tall) → Genotype = 100% Tt, Phenotype = 100% Tall
- **F₂ Generation (Tt × Tt):**
 - Genotypes: TT, Tt, Tt, tt
 - **Genotypic Ratio:** 1 TT : 2 Tt : 1 tt
 - **Phenotypic Ratio:** 3 Tall : 1 Short

This experiment supported Mendel's law that traits segregate independently during gamete formation.

10. In humans, free earlobes (E) are dominant over attached earlobes (e). A person with free earlobes has a child with attached earlobes. What is the genotype of the parent with free earlobes, and what are the possible genotypes of the offspring?

Since the child has attached earlobes (ee), both parents must contribute a recessive allele. This means the parent with free earlobes is **heterozygous (Ee)**, while the other parent is **homozygous recessive (ee)**.

Cross: Ee × ee

Offspring: 50% Ee (free earlobes), 50% ee (attached earlobes).

Thus, there's a 50% chance for each trait to appear in the child.

11. Two pea plants, one with yellow round seeds (YYRR) and another with green wrinkled seeds (yyrr), are crossed. What are the phenotypic ratios in the F₂ generation if the F₁ generation self-fertilizes?

F₁ Generation: All plants will be **YyRr** (yellow and round).

When F₁ plants self-fertilize (YyRr × YyRr), the **F₂ generation** shows:

- **Phenotypic Ratio:** 9 Yellow Round : 3 Yellow Wrinkled : 3 Green Round : 1 Green Wrinkled.
This 9:3:3:1 ratio proved that traits assort independently during gamete formation.

12. In a dihybrid cross, two pea plants with the genotype RrYy are crossed. What is the probability (chance) that an offspring will have round, yellow seeds (RRYY) in the F₂ generation?

Each trait is independent:

- Probability of **RR** = 1/4
- Probability of **YY** = 1/4
- Therefore, probability of **RRYY** = $1/4 \times 1/4 = 1/16$.
So, there is a **1 in 16 chance** that an offspring will have round, yellow seeds with the genotype RRYY.

Section III: Extensive Answer Questions

1. Describe the different parts of a chromosome, including chromatids, centromeres, and telomeres.

A **chromosome** is a thread-like structure found in the nucleus that carries genetic information in the form of DNA. It has several key parts:

- **Chromatids:** Each replicated chromosome has two identical sister chromatids joined together. They contain identical DNA molecules that separate during cell division to ensure each daughter cell gets a copy.
- **Centromere:** The point where two chromatids are attached. It plays a crucial role in attaching chromosomes to spindle fibers during cell division, ensuring equal separation.
- **Telomeres:** Protective caps at each end of the chromosome made of repetitive DNA. They prevent loss of genetic information and protect chromosomes from sticking together.
Together, these structures maintain chromosome stability and ensure accurate transmission of genetic information.

2. Explain how DNA is organized within the chromosome, including the role of histone proteins and nucleosomes.

DNA is an extremely long molecule, so it must be tightly packed inside the nucleus. This organization happens through a combination of **DNA** and **histone proteins**:

- DNA wraps around histone proteins to form small bead-like units called **nucleosomes**.

- These nucleosomes coil further to form **chromatin fibers**, which fold and condense into visible **chromosomes** during cell division.
Histones help in maintaining the shape of DNA, control gene activity, and protect the DNA from damage. This compact structure allows efficient storage and regulation of genetic information inside the nucleus.

3. Describe Mendel's procedure of crossing a true-breeding round-seeded pea plant with a wrinkled-seeded plant. Also, draw the diagram of the cross.

Mendel crossed a **true-breeding round-seeded plant (RR)** with a **wrinkled-seeded plant (rr)**:

- **P Generation:** $RR \times rr$
- **F₁ Generation:** All offspring were **Rr** and had round seeds (dominant trait).
When F₁ plants were self-pollinated ($Rr \times Rr$):
- **F₂ Generation:** Produced offspring with a **3:1 ratio** of round to wrinkled seeds.
This showed that traits are inherited as separate units, not blended.

Diagram:

$RR \times rr \rightarrow F_1$: all Rr (round)

$Rr \times Rr \rightarrow F_2$: 1 RR : 2 Rr : 1 rr $\rightarrow$ Phenotypic ratio 3 Round : 1 Wrinkled

4. Explain the results of the F₁ and F₂ generations in a monohybrid cross and describe the conclusions Mendel drew regarding dominant and recessive traits.

In Mendel's **monohybrid cross**, he studied one pair of contrasting traits — such as seed shape (round vs. wrinkled).

- In the **F₁ generation**, all offspring showed only the **dominant trait** (round seeds).
- In the **F₂ generation**, when F₁ plants were self-pollinated, both traits reappeared in a **3:1 ratio** (three round to one wrinkled).
From these results, Mendel concluded that traits are controlled by **pairs of factors (now called genes)**. One is **dominant** and masks the **recessive** one, but the recessive trait can reappear in the next generation.

5. Define genotype and phenotype, providing examples to illustrate how different genotypes can lead to different phenotypes.

- **Genotype** refers to the genetic makeup or the combination of alleles an organism carries (e.g., TT, Tt, or tt).

- **Phenotype** is the observable trait or physical appearance resulting from the genotype (e.g., tall or short plant).
For example, in pea plants:
- TT and Tt both result in **tall** plants (because T is dominant).
- tt results in **short** plants (recessive).
This shows that different genotypes can produce the same or different phenotypes depending on dominance relationships.

6. Explain what a test cross is and how it can be used to determine whether an organism displaying a dominant trait is homozygous or heterozygous.

A **test cross** is a breeding experiment in which an organism showing a **dominant trait** is crossed with one showing a **recessive trait** (homozygous recessive).

If all offspring show the dominant trait, the parent is **homozygous dominant**.

If some offspring show the recessive trait, the parent is **heterozygous**.

For example, if a tall pea plant (unknown genotype) is crossed with a short plant (tt):

- All tall offspring → parent is **TT**.
- Tall and short offspring → parent is **Tt**.
Thus, a test cross helps in determining the exact genotype of an organism.